



Determinants of Financial Literacy Among Italian Adults: The Role of Demographic, Educational, and Digital Factors

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Team Members	Matricola	Email
Abdul Sami	939542	a.sami7@campus.unimib.it
Davina David	946851	d.david2@campus.unimib.it

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Keywords

Financial literacy, financial behavior, financial attitudes, Italy, digital finance

About Dataset

The dataset used for this project is a **Survey on Surveys on Financial Literacy and Digital Financial Skills in Italy (Adults) IACOFI 2023** Microdata, distributed by the Bank of Italy. The project specifically uses the file `database_eng.dta` in Stata (.dta) format.

Declaration on AI Tools

We used AI tools **Perplexity** during the preparation of this report to support the following tasks:

- Structuring the outline of the report in accordance with the exam instructions;
- Improving the clarity and wording of selected sections;
- Checking the coherence of methodological descriptions.

All data loading, cleaning, scoring, analysis, and interpretation of results were performed independently by the group, based on our own understanding of the dataset, the OECD/INFE methodology, and the course material.

1. Abstract

This study investigates the demographic, educational, and digital finance-related factors that explain differences in financial literacy among Italian adults. The analysis is based on the 2023 IACOFI survey microdata from the Bank of Italy, covering approximately **5000 individuals aged 18–79** and measuring three dimensions of financial literacy: knowledge, behaviour, and attitudes. A composite financial literacy score and a digital finance score are constructed to support the analysis. The study employs descriptive statistics and multiple linear regression to assess the impact of **age, gender, region, education, income**, and online banking behaviour on financial literacy outcomes. The results indicate that education and selected digital financial behaviours are significant predictors of financial literacy, while gender and region have smaller but still measurable effects. Basically, the study identifies population groups at risk of lower financial literacy and demonstrates a reproducible data science workflow for survey-based analysis.

2. Research Question

2.1 Statement

2.1.1 Main Research Question

To what extent do demographic, educational, and digital finance-related factors explain differences in financial literacy among Italian adults?

2.1.2 Sub-Questions

- RQ1: Does education level influence financial literacy?
- RQ2: Are there demographic differences in financial literacy (**age, gender, region**)?
- RQ3: Is digital finance behaviour associated with higher financial literacy?
- RQ4: Which factors are the strongest predictors of financial literacy?

In this study, *financial literacy* is defined as a composite score based on three dimensions:

- **Knowledge:** objective understanding of financial concepts such as inflation, interest rates, and risk diversification.
- **Behaviour:** actual financial practices such as budgeting, saving, and financial planning.
- **Attitudes:** preferences and orientations toward saving, monitoring finances, and long-term planning.

Digital finance behaviour refers to the frequency with which individuals use digital tools to manage their finances, including activities such as checking account balances, paying bills, transferring money, and managing financial products online.

2.2 Motivation and Stakeholders

Understanding financial literacy in Italy is not only a matter of measuring financial knowledge, but also of identifying who is left behind and why. Italian adults differ significantly in their access to education, exposure to digital financial tools, and socioeconomic conditions. These factors do not operate in isolation: for example, individuals with lower education levels, living in Southern Italy, and with limited engagement in digital finance may face overlapping disadvantages that a simple descriptive analysis would fail to capture. By examining how demographic, educational, and digital finance-related factors jointly explain financial literacy, this study provides a more comprehensive picture of financial vulnerability among Italian adults.

This research is relevant to several stakeholders. Policymakers and financial education authorities, such as the Bank of Italy and OECD/INFE, can use these findings to identify the main drivers of financial literacy gaps and design more targeted interventions. Financial institutions and educators developing digital finance tools may benefit from understanding whether digital engagement is associated with higher financial literacy or instead concentrated among already advantaged groups. More broadly, this project demonstrates how data science methods taught in the course can be applied to real-world survey data to generate policy-relevant insights rather than purely descriptive statistical outputs.

3. About Data

3.1 Data Description

The dataset used in this study is the 2023 IACOFI survey (*Surveys on Financial Literacy and Digital Financial Skills in Italy: Adults*), collected by the Bank of Italy through Computer-Assisted Telephone Interviewing (CATI) on a representative sample of approximately 5000 adults aged 18–79.

The variables used in the analysis are grouped into three main categories:

1. Demographics and socio-economic background: This group includes **age (qd7)**, **gender (qd1; 1 = male, 0 = female)**, **region of residence (qd2; North West, North East, Centre, South, Islands)**, **urban/rural location (qd3)**, household composition, **employment status (qd10)**, **education level (qd9)**, and **income bracket (qd13)**.

2. Financial literacy components

Financial literacy is measured through three dimensions:

- **Knowledge:** Objective questions assessing understanding of financial concepts such as inflation, simple and compound interest, and risk diversification (qk3, qk4, qk5, qk6, qk10).
- **Behaviour:** Indicators of day-to-day financial management and planning, such as budgeting, saving in deposit accounts, and setting up automatic payments (qf2_*, qs2_3, qs2_5).
- **Attitudes:** Statements capturing financial mindset, including attention to personal finances and long-term planning orientation (qs1_1, qs1_5, qs1_8).

3. Digital finance behaviour: This category captures the frequency of online financial activities (qp9_1 to qp9_6), including checking account balances, paying bills, purchasing goods, transferring money, and managing financial products online.

All responses are recorded on ordinal scales (1–5 ranging from “Never” to “Very often”). Missing or non-response values are coded using standard survey conventions (-99, -98, -97, -999).

3.2 Strength and Weakness of Data

3.2.1 Strengths

- The dataset is based on a nationally representative sample of Italian adults, which allows for generalization of results within the Italian adult population.
- It provides a rich and multidimensional measure of financial literacy, covering knowledge, behaviour, and attitudes, together with digital finance behaviour, and is aligned with the OECD/INFE framework.
- The microdata are publicly available in standard formats (**Stata**, **CSV**), ensuring transparency and reproducibility of the analysis.

3.2.2 Weaknesses

- The cross-sectional design captures individuals at a single point in time, which prevents causal inference (whether education leads to higher financial literacy or vice versa).
- The use of self-reported measures for behaviour and attitudes may introduce bias, such as social desirability bias or misreporting.
- Several variables are recorded in grouped formats (income and age bands), which limits the precision of continuous or more granular statistical analysis.

Despite these limitations, the dataset is well-suited to addressing the research questions and provides sufficient depth for an empirical analysis of financial literacy in Italy.

4. Methodology

4.1 Basically Logic thread of the Study

The analysis follows the steps below:

1. Load and clean the dataset in Python, including handling missing values and verifying data types.
2. Define and prepare variables by renaming relevant columns and recoding ordinal scales where necessary.
3. Construct composite indices for financial literacy components (knowledge, behaviour, and attitudes) as well as a digital finance score, following the OECD/INFE framework.
4. Explore the data by examining distributions and correlations of the composite scores with demographic and digital finance variables using descriptive statistics and visualizations.
5. Estimate a multiple linear regression model to explain variation in the Basically financial literacy score using demographic, educational, and digital finance variables.
6. Interpret the results and discuss key insights and limitations, particularly in relation to the cross-sectional nature of the survey and the use of categorical variables.

The Basically objective is to ensure a clear and traceable analytical workflow, starting from a well-defined research question and maintaining interpretability for non-technical stakeholders.

4.2 Computational Tools and Methods

- **Python (pandas, numpy, matplotlib, seaborn):**

Python is used as the main programming language due to its widespread adoption in data science and its strong ecosystem for data manipulation, analysis, and visualization. In particular, pandas and numpy are used for data processing, while matplotlib and seaborn support reproducible and informative visualizations.

- **Linear regression (sklearn.linear_model.LinearRegression):**

A linear regression model is employed to analyse the relationship between financial literacy and its potential determinants. This method is chosen because the dependent variable is a continuous composite score, and linear regression provides interpretable coefficients that allow the relative importance of predictors to be assessed.

- **Descriptive statistics and Visualizations:**

Summary statistics and graphical methods (including bar plots, boxplots, density plots, and correlation heatmaps) are used to explore the data structure and to complement the regression analysis with intuitive visual insights.

All analyses are implemented in a Jupyter Notebook or Python script. The corresponding code is provided in the submission.

5. Results

5.1 Methodological Workflow of the Study

The analysis began by loading the file *database_eng.dta* into Python and cleaning the dataset. Invalid or non-response codes (**-99, -98, -97, and -999**) were recoded as missing values. The sample was then restricted to respondents with complete information for all variables used in the analysis, ensuring a consistent working dataset for subsequent steps.

Next, composite indicators were constructed to measure the main dimensions of financial literacy. The **financial knowledge score** was based on objective questions related to inflation, interest rates, compound interest, and other core financial concepts. The **financial behaviour score** captured budgeting, saving, and financial planning practices, while the **financial attitude score** reflected respondents' orientation toward long-term planning and financial monitoring. These three components were then combined to form an Basically **financial literacy score**.

Additionally, a **digital finance score** was created using items measuring online financial activities such as checking account balances, paying bills, transferring money, and managing financial products online. Higher values of this score indicate more frequent use of digital financial services.

The dataset was then explored using descriptive statistics and visual tools, including boxplots, scatter plots, a correlation heatmap, and a histogram of the main score distribution. **K-means clustering** was subsequently applied to identify broad respondent profiles, with the number of clusters selected using the **elbow** method ($k = 3$). A PCA projection of the clustered data revealed three partially overlapping groups, suggesting a gradual transition in financial capability rather than clearly separated categories.

Finally, a **linear regression model** was estimated to examine the relationship between financial literacy and its determinants, including **education, age, gender, region, income, and digital finance** behaviour. The model was used to identify the variables most strongly associated with the Basically financial literacy score.

5.2 Result and Interpretations

5.2.1 Distribution of Financial Literacy Scores

We use a histogram to explore the distribution of the Basically financial literacy score and to assess its central tendency and variability (Figure 5.2.1).

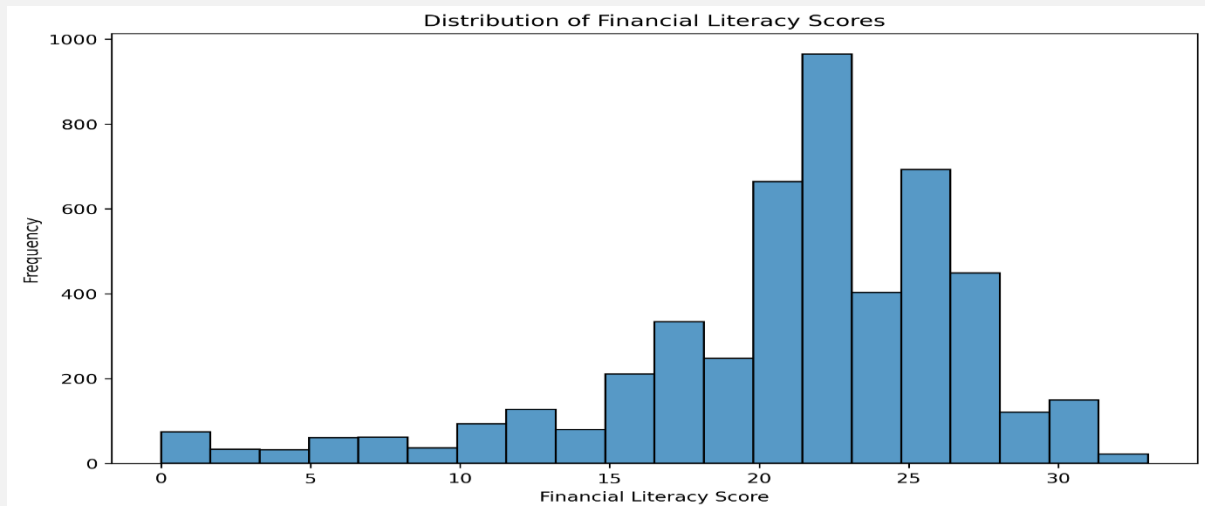


Figure 5.2.1: Distribution of the Basically financial literacy score

Interpretation:

The histogram of financial literacy scores shows that most respondents are concentrated around the middle of the distribution, with the largest density of observations between approximately **20 and 27**. This suggests that financial literacy among Italian adults is generally moderate, rather than being concentrated at very low or very high levels. At the same time, a smaller group of respondents records very low scores (approximately 0–10), indicating that financial vulnerability remains present within the sample.

This moderate concentration of scores provides a useful context for analyzing how demographic and educational factors are distributed across different levels of financial literacy.

5.2.2 Financial Literacy by Gender

We use a boxplot to compare the distribution of financial literacy scores across gender groups

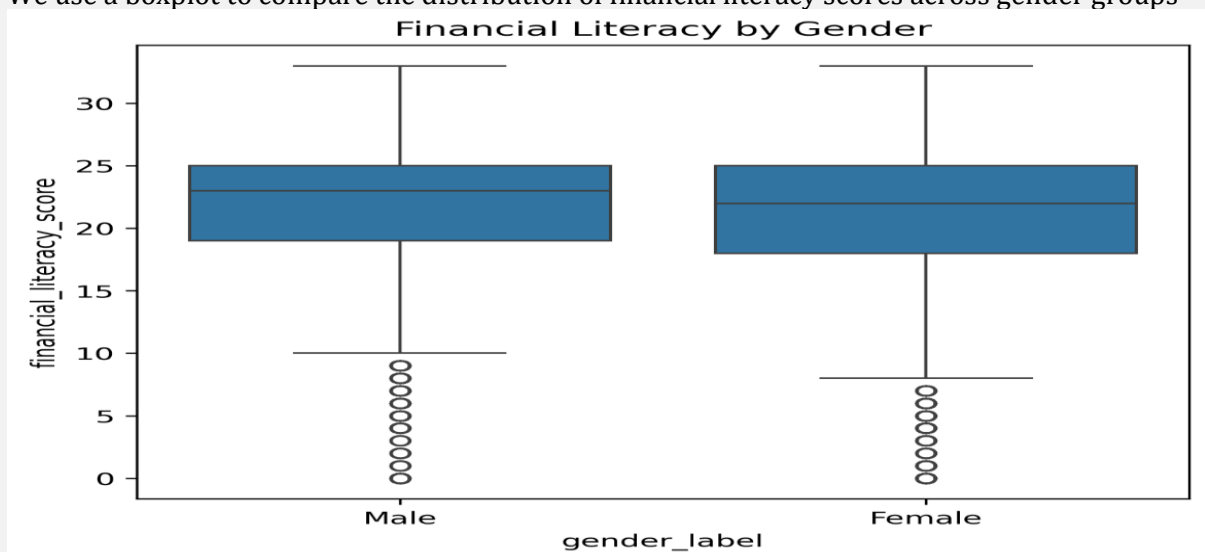


Figure 5.2.2: Financial Literacy by Gender

Interpretation:

The boxplot by gender shows only small differences between men and women. Male respondents have a median financial literacy score of approximately **23**, while female respondents have a median of approximately **22**. The interquartile ranges are also very similar, roughly between **19 and 25 for men and 18 and 25 for women**. This indicates that gender differences exist but are limited, with substantial overlap between the two distributions.

In simple gender alone does not appear to be a strong determinant of financial literacy, it is useful to consider it alongside other factors such as education and age to examine potential interactions.

5.2.3 Financial Literacy by Education Level

We use a boxplot to examine how financial literacy scores vary across different levels of education (figure 5.2.3)

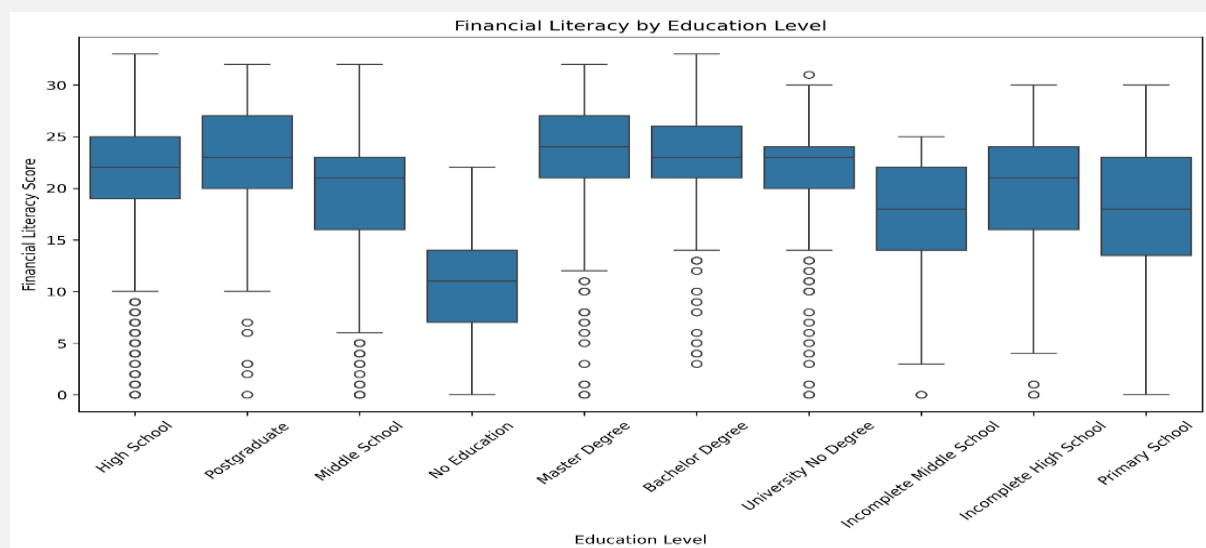


Figure 5.2.3: Financial Literacy by Education Level

Interpretation:

The boxplot by education level reveals a clear and strong gradient. Respondents with master's or postgraduate education have median **financial literacy scores** of approximately **24**, while those with a **bachelor's degree score** around **23**. Individuals with a high school education have a **lower median of about 22**. In contrast, respondents with primary education or no formal education exhibit substantially lower median scores, around **18 and 11** respectively.

This pattern confirms that education is one of the strongest predictors of financial literacy in the sample. Basically, the strong educational gradient represents the most consistent finding in the dataset and provides a key reference point for interpreting the effects of other variables.

5.2.4 Financial Literacy by Age Group

We use a boxplot to analyse how financial literacy scores vary across different age groups (Figure 5.2.4).

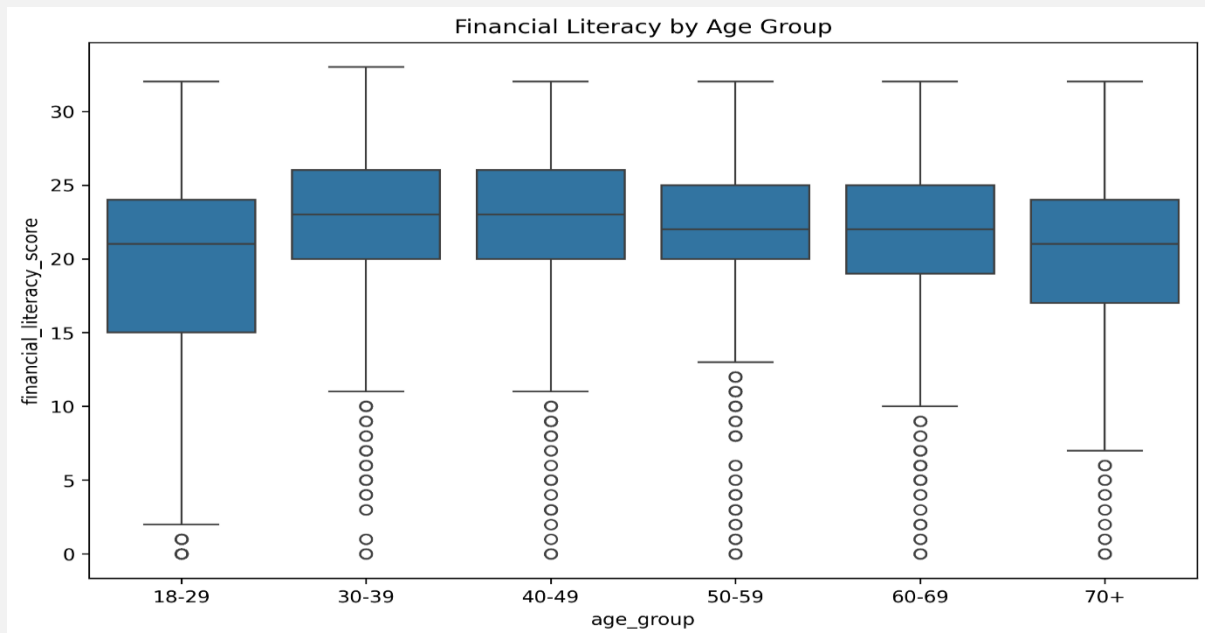


Figure 5.2.4: Financial Literacy by Age Group

Interpretation:

The boxplot by age group suggests that **financial literacy is highest among adults aged 30–39 and 40–49**, with median scores of approximately 23. The **18–29** group shows a lower median and a wider distribution, while **older age groups such as 60–69 and 70+ exhibit a slight decline in median scores**.

The pattern suggests that **financial literacy increases with age** and experience up to mid-adulthood, before declining slightly in later life. This inverted-U pattern complements the findings on education, indicating that both formal education and life experience play important roles in shaping financial capability.

5.2.4 Relationship between Digital Finance Behaviour and Financial Literacy

We use a scatter plot to examine the relationship between digital finance behaviour and financial literacy (Figure 5.2.4)

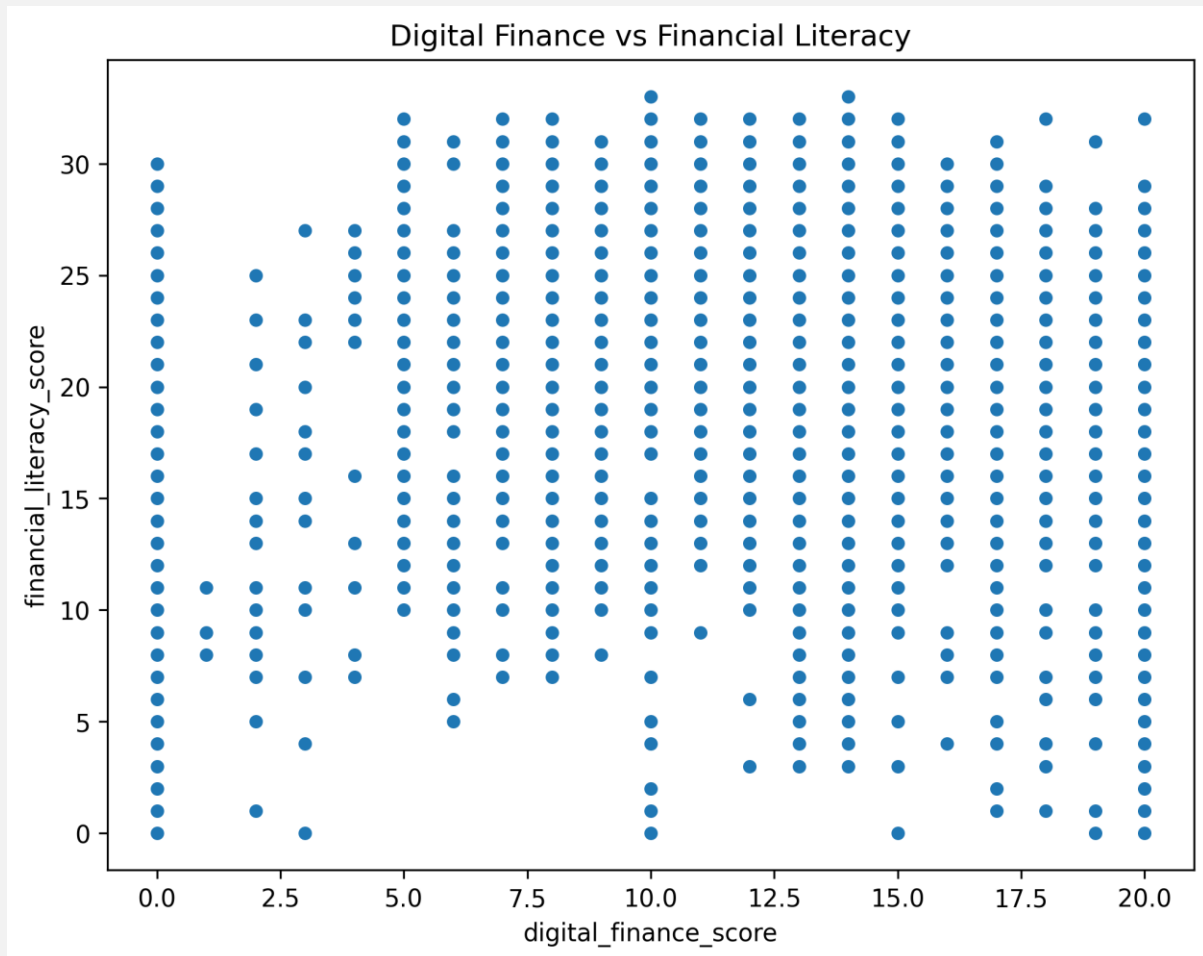


Figure 5.2.4: Digital Finance vs Financial Literacy

Interpretation:

The scatter plot of **digital finance score** versus financial literacy indicates a **positive but weak** relationship. The **correlation** is approximately **0.17**, suggesting that individuals who use digital financial tools more frequently tend to have slightly higher financial literacy scores. However, the relationship is not strong, indicating that digital finance usage explains only a small proportion of the variation in financial literacy.

Basically, this weak but positive association suggests that digital finance engagement is part of the broader financial literacy profile, but it acts more as a complement to education rather than a substitute for it.

5.2.5 Correlation Analysis of Financial Literacy Components

We use a correlation heatmap to examine the strength and direction of the relationships between financial literacy, its components, and digital finance behaviour (Figure 5.2.5).

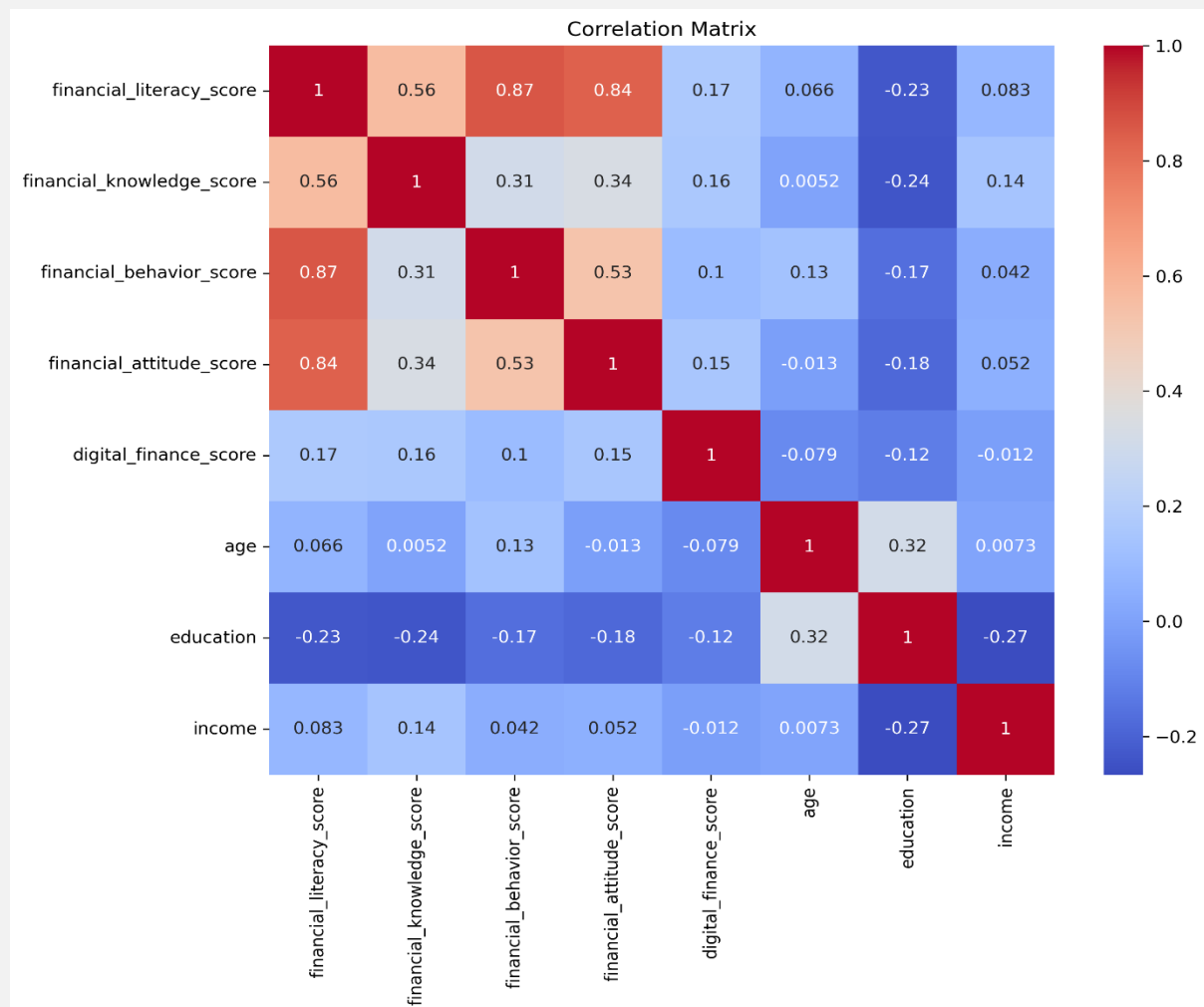


Figure 5.2.5: Correlation Matrix of Financial Literacy Components

Interpretation:

The **correlation heatmap** highlights the strongest associations within the dataset. In simple financial literacy is strongly correlated with financial behaviour (**0.87**) and financial attitudes (**0.84**), indicating that these dimensions are closely linked to the composite literacy score. In contrast, **the relationship between financial literacy and financial knowledge is more moderate (0.56)**. **The correlation between financial literacy and digital finance score is weaker**, at approximately **0.17**, suggesting that digital finance behaviour contributes to financial literacy but explains only a limited share of its variation.

The sign of the education correlation depends on the coding scheme used in the dataset, where lower numeric values correspond to higher levels of educational attainment.

Basically, the heatmap confirms that financial behaviour and attitudes are the dimensions most strongly associated with Basic financial literacy, while financial knowledge and digital finance behaviour play a more supportive role.

5.2.6 PCA Cluster Analysis of Financial Literacy Profiles

We use Principal Component Analysis (PCA) combined with K-means clustering to identify broad respondent profiles based on financial literacy and digital finance characteristics (Figure 5.2.6).

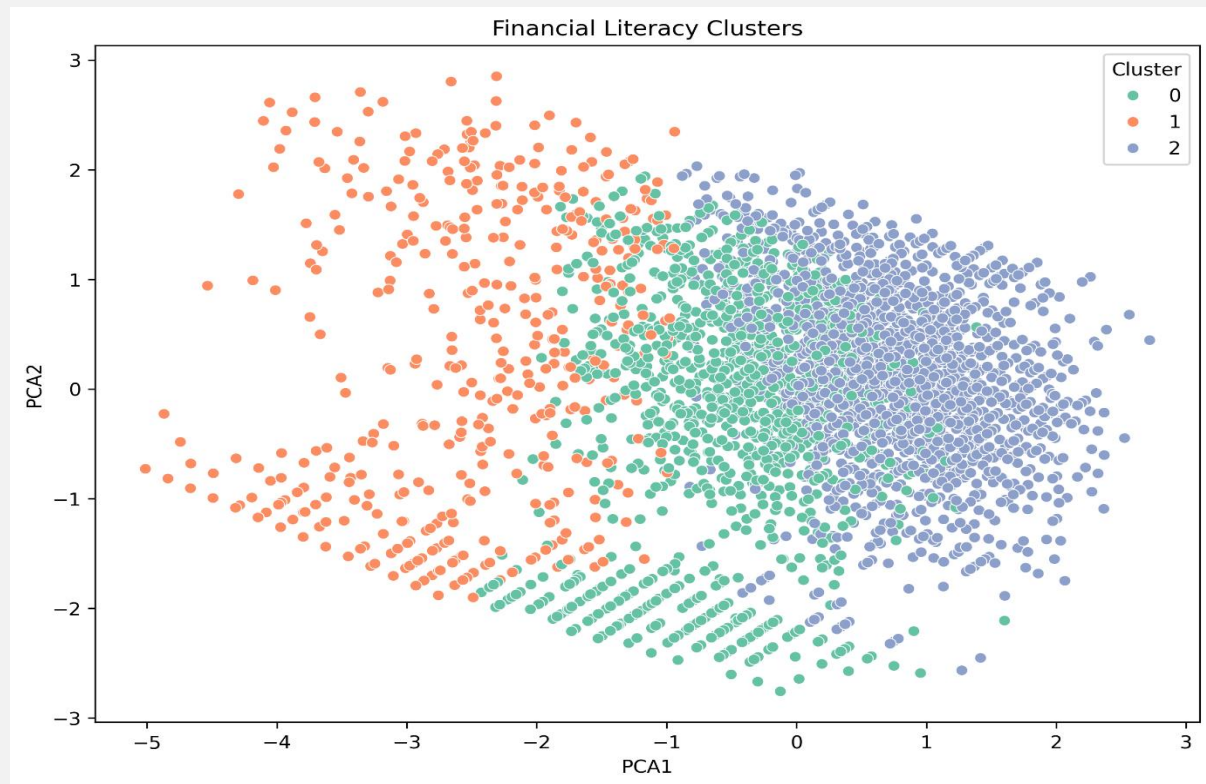


Figure 5.2.6: Financial Literacy Clusters using PCA and K-Means Clustering

Interpretation:

The PCA visualization identifies three distinct financial literacy clusters using K-means clustering (**K= 3**). Cluster 1 (**orange**) represents respondents with relatively **lower financial literacy** and digital finance scores, while Cluster 2 (**blue**) includes individuals with **stronger financial** behaviour, attitudes, and digital engagement. Cluster 0 (green) appears to represent respondents with **moderate financial literacy** characteristics.

Although the clusters are distinguishable, some overlap exists between groups, indicating a **gradual transition from lower to higher financial literacy profiles rather than sharply separated categories**. Basically, the results demonstrate meaningful variation in financial literacy patterns among Italian adults.

The gradual transitions visible in the PCA plot suggest that financial literacy profiles are shaped by combinations of demographic, educational, behavioural, and digital finance factors rather than by a single dominant variable.

5.2.7 Elbow Method for Cluster Selection

We use the elbow method to determine the optimal number of clusters for the K-means clustering analysis (Figure 5.2.7).

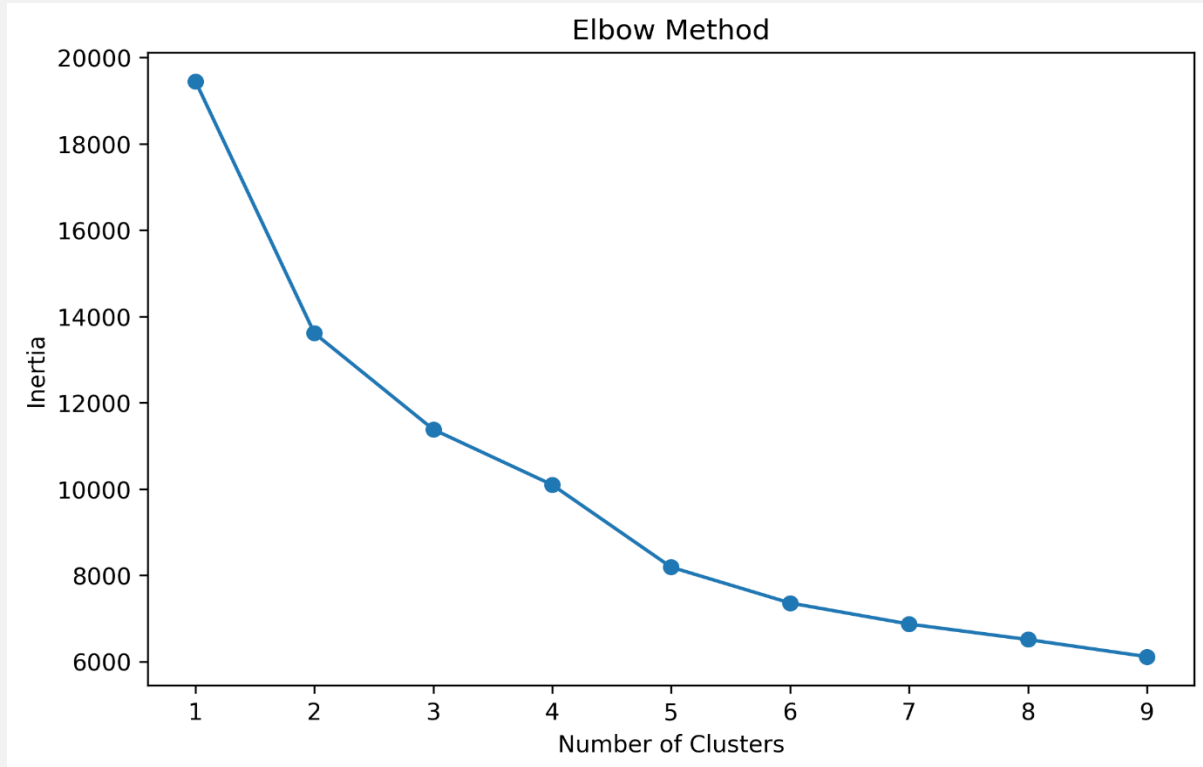


Figure 5.2.6: Elbow Method for Selecting the Optimal Number of Clusters

Interpretation:

The **elbow method** suggests that **three clusters** provide the most appropriate solution for the **K-means analysis**. **Inertia decreases sharply from approximately 19,500 at (K = 1) to around 13,600 at (K = 2), and then to nearly 11,400 at (K = 3).** Beyond this point, the reduction in inertia becomes considerably smaller, indicating diminishing improvements in model fit as additional clusters are added.

This pattern suggests that a **three-cluster solution achieves a suitable balance between simplicity and explanatory power**. The PCA cluster visualization further supports this conclusion by revealing three broad groups with partial overlap, indicating a gradual transition from lower to higher financial capability rather than completely distinct categories.

Basically, the three-cluster solution provides a practical and interpretable segmentation of the Italian adult population while avoiding unnecessary complexity or overfitting.

6. Conclusion

6.1 Gap Between Research Objectives and Results

Our goal was to understand which demographic, educational, and digital finance-related factors explain differences in financial literacy among Italian adults, and to do so in a way that produces an integrated interpretation rather than a set of isolated answers. While the analysis partially achieves this objective, an important gap remains between the ambition of fully explaining financial literacy and what the empirical results can actually support.

Education emerges as the strongest individual predictor, but its effect is best understood alongside age, regional context, and digital finance engagement, all of which contribute to a broader profile of financial capability. The regression framework quantifies these relationships in a unified model, and the clustering analysis confirms that Italian adults are distributed along a continuous gradient of financial literacy rather than being divided into sharply separate groups. These findings provide a coherent descriptive and associative picture of financial literacy in Italy.

However, the gap lies in the fact that these results remain fundamentally correlational. The cross-sectional nature of the data does not allow causal interpretation, meaning that the analysis cannot fully explain the underlying mechanisms driving financial literacy differences. As a result, while the study identifies strong and meaningful associations and an integrated structure of financial capability, it does not fully achieve the objective of explaining causal drivers of financial literacy.

Basically, the findings suggest that financial vulnerability in Italy is concentrated among individuals with lower education, limited digital engagement, and specific demographic profiles, and that effective policy interventions should address these factors in combination rather than in isolation, while acknowledging the limits of causal inference.

6.2 Strength and Weakness of the Study

Strengths

- The use of a nationally representative dataset ensures that the findings are broadly generalizable to the Italian adult population and are aligned with OECD/INFE standards.
- The construction of financial literacy and digital finance scores is transparent and reproducible, supported by clearly documented code.
- The analysis provides an integrated interpretation of results, combining demographic, educational, and digital finance factors into a coherent analytical framework.

Weaknesses

- The cross-sectional design of the dataset limits the ability to draw causal inferences.
- The use of grouped variables (income and education bands), together with self-reported measures, reduces the precision of the analysis and may introduce measurement bias.
- The dataset does not capture potentially important omitted factors such as early-life financial education, cultural attitudes, or previous financial shocks.

6.3 Directions for Further Research and Improvements

Future research could extend this study by using longitudinal data from subsequent IACOFI waves, enabling panel-based methods that better identify causal relationships over time. Additionally, estimating separate regression models for different geographical areas (**Northern and Southern Italy**) or age cohorts could help determine whether the determinants of financial literacy vary across population subgroups.

Experimental or quasi-experimental approaches could also be used to assess the impact of financial education interventions on both financial literacy and digital finance adoption. Finally, the inclusion of more detailed income, wealth, and household composition variables would allow for a more comprehensive analysis of financial resilience and economic wellbeing.

7. References

1. Bank of Italy. (2024). *Financial literacy of Italian adults – IACOFI 2023 Microdata*. Available at: [link](#)
2. OECD, **OECD/INFE 2023 International Survey of Adult Financial Literacy**:. [Link](#)
3. Bank of Italy page for the 2023 adult survey and Italian results [Link](#)